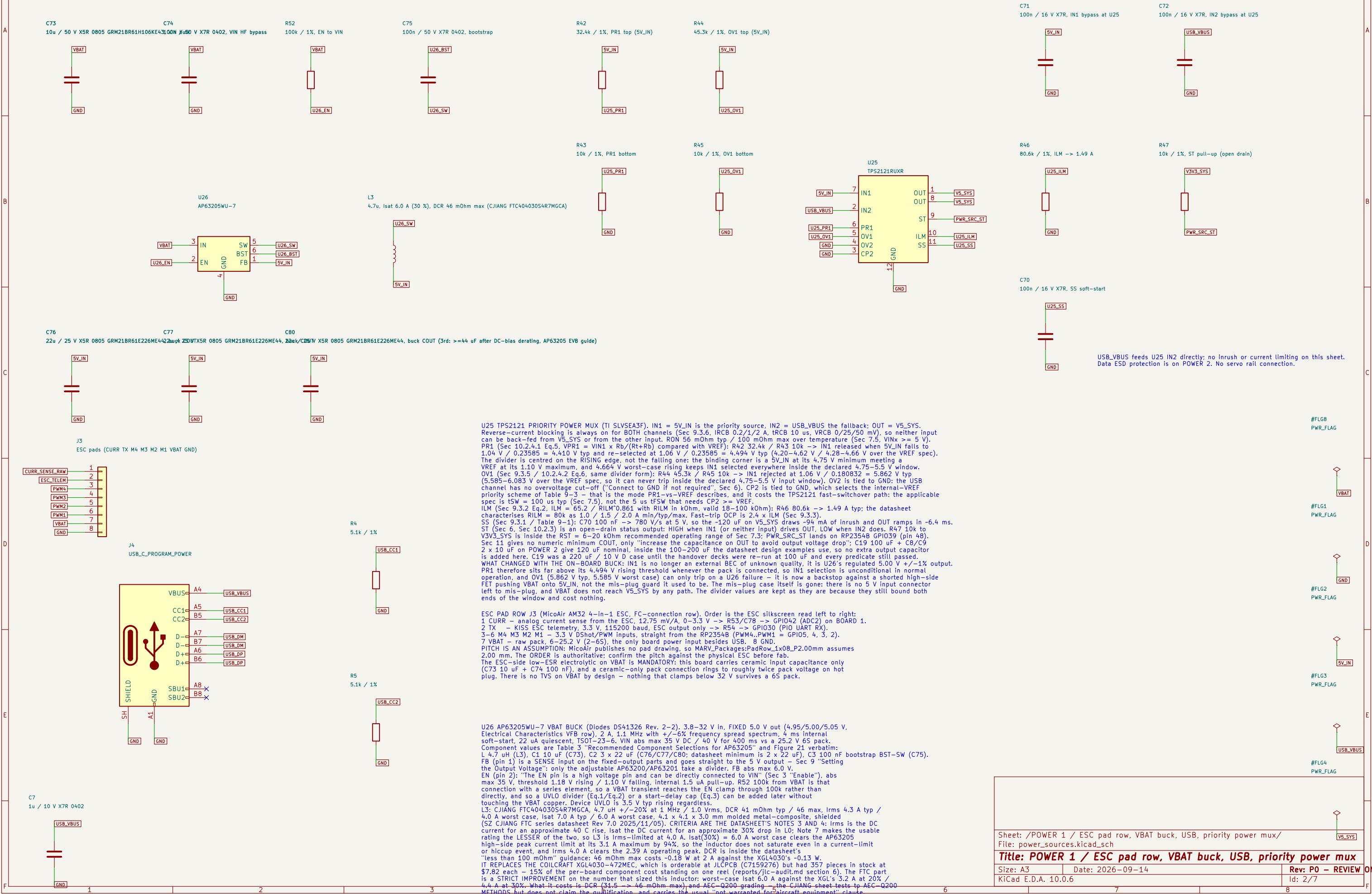


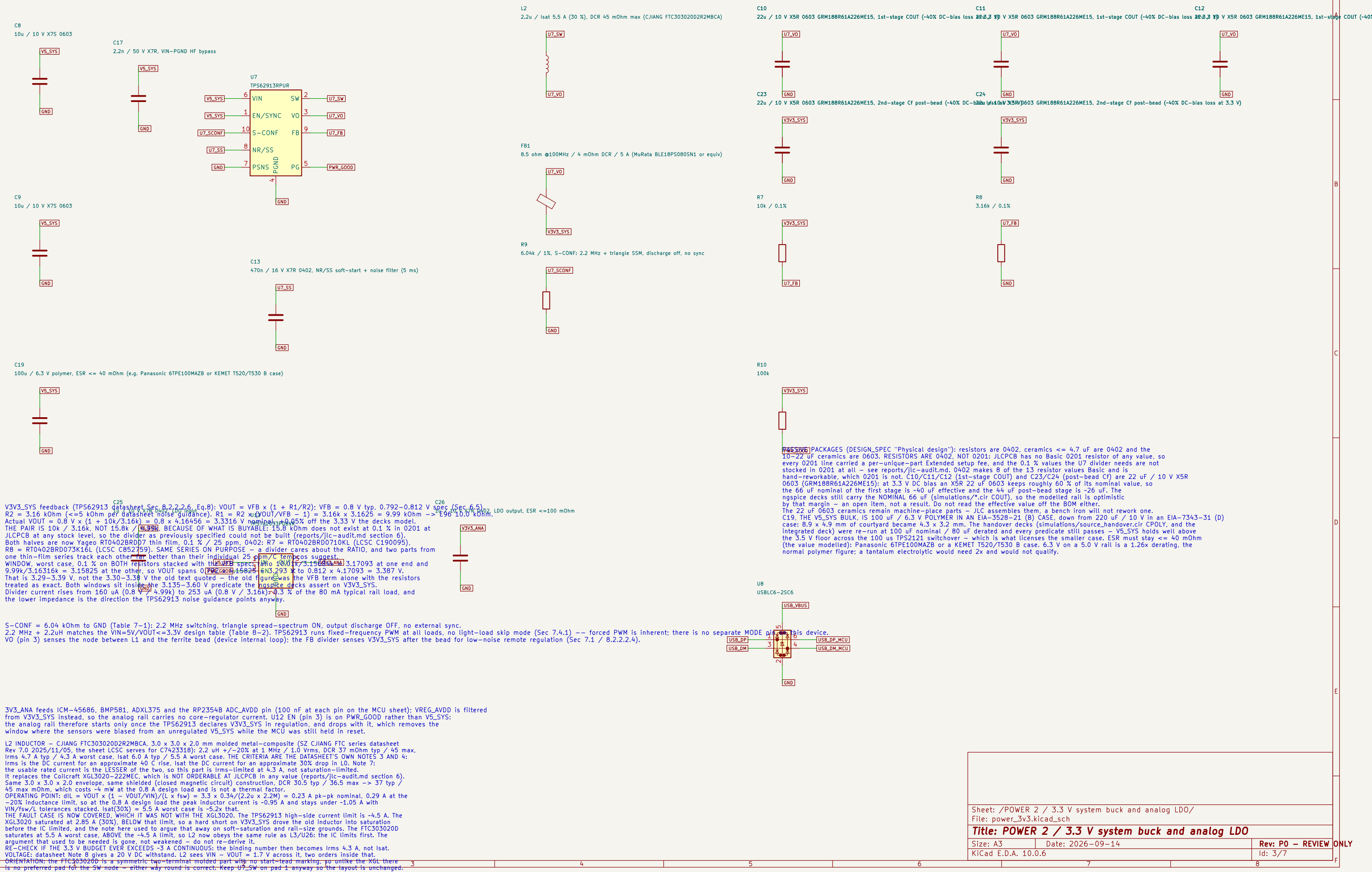
	1	2	3	4	5	6	7	8
A	Integrated draft: power chain (sheets 2–4) and board stage (sheets 5–8). See DESIGN_SPEC.md. Inputs: VBAT 6–25.2 V (2–6S) from the MicoAir AM32 ESC pad row J3, and USB VBUS. Rails: 5V_IN from the U26 AP63205 buck, V5_SYS from the U25 TPS2121 mux, V3V3_SYS from the TPS62913 buck, V3V3_ANA from the TPS7A20 LDO. Servo 5 V is 5V_IN, upstream of the mux. No PCB placement or routing has been performed.							
B	POWER 1 / ESC pad row, VBAT buck, USB, priority power mux File: power_sources.kicad_sch				BOARD 2 / IMU, barometer, high-g File: sensors.kicad_sch			
C	POWER 2 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch				BOARD 3 / log flash and microSD File: storage.kicad_sch			
D	POWER 3 / IO array File: power_periph.kicad_sch							
E	BOARD 1 / RP2354B File: mcu.kicad_sch							
F	1	2	3	4	5	6	7	8

Sheet: / File: MARV-V2.kicad_sch		
Title: MARV integrated power – preliminary		
Size: A3	Date: 2026-09-14	Rev: P0 – NOT FOR FAB
KiCad E.D.A. 10.0.6	Id: 1/7	

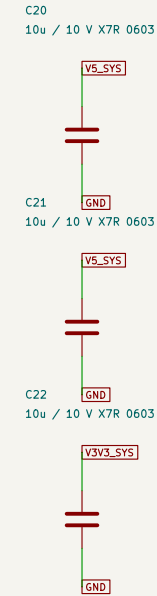
BOARD INPUTS: VBAT 6–25.2 V (2–6S) from the ESC pad row J3, and USB VBUS ~5 V (J4). U26 (AP63205WU) makes 5V_IN from VBAT; 5V_IN is the priority input of the U25 power mux and also the servo 5 V bus.



AVIONICS ONLY: 300 mA continuous / 500 mA short peak, provisional. NO SERVO POWER.

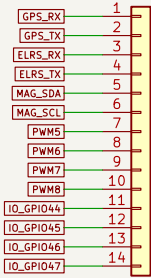


IO ARRAY – ONE 3-COLUMN x 14-ROW THROUGH-HOLE BLOCK ON THE RIGHT EDGE, 2.54 mm, madflitght FC3v2 style. It replaces the old 2x16 left-edge array AND the J12/J13/J14 servo block: every signal that leaves this board except the ESC row (J3) and the SWD pads (J10) now leaves on this one block. COLUMN RULE, from the board edge inward: GND (J8) | POWER (J7) | SIGNAL (J6). The SIGNAL column is the INNERMOST one, next to the MCU, so the 14 signal traces are the short ones; the two rails run down the outside where a plane stitch is cheap. A 3-wire module lead (signal, power, ground) plugs straight across ONE row. THREE Conn_01x14 SYMBOLS, not one 3x14: no 3-row generic connector symbol ships with this KiCad install, and three 1x14 rows with a body-trimmed courtyard abut exactly on the 2.54 mm grid (see tools/setup_pcb.py). DS-009 NAMING IS KEPT: a port label is named for the PERIPHERAL pin it belongs to, so GPS_RX / ELRS_RX are module inputs driven by the RP2354B UART TX pins (rows 1 and 3, silk T0 / T1) and GPS_TX / ELRS_TX are module outputs that land on the MCU UART RX pins (rows 2 and 4, silk R0 / R1).

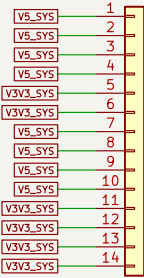


C20 / C21: V5_SYS local bulk at the array (rows 1–4 GPS/ELRS and rows 7–10 servo, the 5 V supply rows). C22: V3V3_SYS local bulk at the array (rows 5–6 and 11–14).

J6
IO array signal column (innermost)



J7
IO array power column (5V / 3V3 per row)



J8
IO array GND column (board edge)



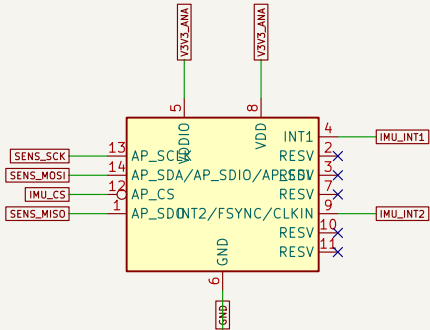
IO ARRAY ROW MAP (row n = J6 pin n signal J7 pin n power J8 pin n GND)			
1 T0	GPS_RX	V5_SYS	GND UART0 TX -> GPS RX (GPIO12)
2 R0	GPS_TX	V5_SYS	GND UART0 RX <- GPS TX (GPIO13)
3 T1	ELRS_RX	V5_SYS	GND UART1 TX -> ELRS RX (GPIO8)
4 R1	ELRS_TX	V5_SYS	GND UART1 RX <- ELRS TX (GPIO9)
5 SDA	MAG_SDA	V3V3_SYS	GND I2C1 SDA (GPIO6), 4.7k pull-up on the MCU sheet
6 SCL	MAG_SCL	V3V3_SYS	GND I2C1 SCL (GPIO7), 4.7k pull-up on the MCU sheet
7 S5	PWM5	V5_SYS	GND PWM5 A (GPIO10), servo 5
8 S6	PWM6	V5_SYS	GND PWM5 B (GPIO11), servo 6
9 S7	PWM7	V5_SYS	GND PWM7 A (GPIO14), servo 7
10 S8	PWM8	V5_SYS	GND PWM7 B (GPIO15), servo 8
11 A44	IO_GPI044	V3V3_SYS	GND GPIO44 = ADC4, spare
12 A45	IO_GPI045	V3V3_SYS	GND GPIO45 = ADC5, spare
13 A46	IO_GPI046	V3V3_SYS	GND GPIO46 = ADC6, spare
14 A47	IO_GPI047	V3V3_SYS	GND GPIO47 = ADC7, spare

PWM MAP
PWM5 = GPIO10 (row 7, S5)
PWM6 = GPIO11 (row 8, S6)
PWM7 = GPIO14 (row 9, S7)
PWM8 = GPIO15 (row 10, S8)

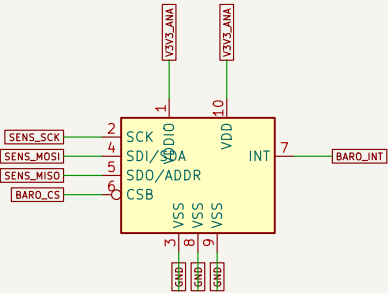
ON THE ESC PAD ROW (J3, POWER 1)
PWM1 = GPIO2
PWM2 = GPIO3
PWM3 = GPIO4
PWM4 = GPIO5

RAILS ON THE ARRAY: rows 1–4 (GPS, ELRS) and rows 7–10 (servos S5–S8) take V5_SYS; rows 5–6 (magnetometer) and rows 11–14 (the four exposed spares) take V3V3_SYS. Every row has its own GND pin in the J8 column, so a signal return is never more than 2.54 mm away and no row can be mis-plugged into a neighbour's ground. SERVO 5 V NOW COMES FROM V5_SYS, NOT 5V_IN. The old J13 servo row tapped 5V_IN directly at the U26 buck output, ahead of the U25 priority mux, so servo current was outside the mux current limit. With one power column the array cannot carry two different 5 V nets under one "5V" silk label, so all eight 5 V rows are V5_SYS: servo current now crosses U25 and counts against its ILM limit (1.49 A, R46) and against the U26 buck. Budget servo current against BOTH – see DESIGN_SPEC "Rails and load budget". CURRENT: rows 1–4 and 7–10 count against the U26 buck through the mux; rows 5–6 and 11–14 count in the V3V3_SYS budget. No fuse and no per-row current limit: the array is a system-integration connector, not a protected port.

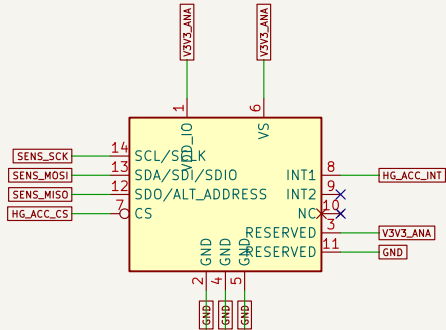
All three sensors run 4-wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TP57A20 LDO). Every supply pin on this sheet gets 100 nF + 1 uF; BMP581 and ADXL375 because their own datasheets call for the pair, the ICM-45686 as a deliberate addition – its datasheet BOM (DS-000577 Rev 1.0 Table 11) lists 0.1 uF only.



U21
ICM-45686 6-axis IMU



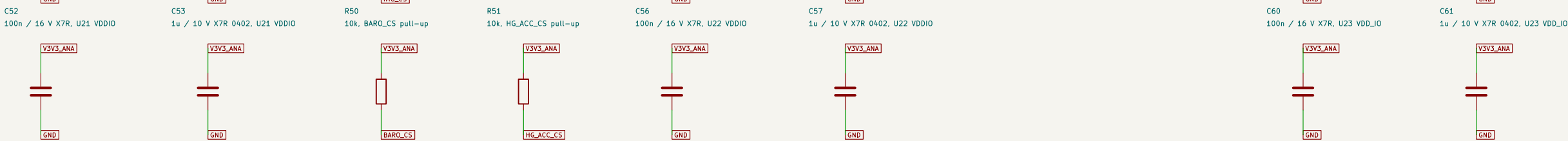
U22
BMP581 barometer



U23
ADXL375 high-g accelerometer



TEST POINTS TP1-TP10 (TestPoint:TestPoint_Pad_1.0x1.6mm, 1 x 1 mm bare F.Cu pads, 50 model by design – tools/audit footprints by --no-3d-ok covers TestPoint, along with the pad rows and mounting holes). TP1 SENS_SCK | TP2 SENS_MOSI | TP3 SENS_MISO | TP4 IMU_CS | TP5 BARO_CS | TP6 HG_ACC_CS | TP7 IMU_INT1 | TP8 IMU_INT2 | TP9 BARO_INT | TP10 HG_ACC_INT. Placement: Inside the sensor cluster, within 6 mm of U21/U22/U23 (tools/setup_pcb.py checks it). They add ten stubs to the SPI bus, which is why they are 1 mm pads next to the parts rather than a header somewhere else: at 10 MHz SCK a short stub is a capacitive load, a long one is a transmission line. No series resistors and no ground pad of their own – probe ground comes off a mounting hole or the J6 array.



U21 ICM-45686 (TDK InvenSense DS-000577 Rev 1.0, Table 10 "Signal Descriptions" / Figure 4 pin-out, Single Interface SPI mode, Figure 10): AP_CS (pin 12) = IMU_CS, AP_SCLK (13) = SENS_SCK, AP_SDA/AP_SDIO/AP_SDI (14) = SENS_MOSI, AP_SDO/AP_ADO (1) = SENS_MISO, INT1 (4) = IMU_INT1, INT2/FSYNC/CLKIN (9) = IMU_INT2 – typed bidirectional because FSNC/CLKIN are host-driven inputs while INT2 is a chip output; only the INT2 function is used here. VDD (8) and VDDIO (5) are both on V3V3_ANA; GND (6) is the only ground pin. Pins 2, 3, 7, 10 and 11 (RESV) are all left No Connect; Table 10 allows No Connect / VDDIO / GND for every RESV pin, and the Single-Interface SPI typical application schematic (Figure 10) shows all five left unterminated (internal pull enabled by default, per the per-pin notes in Table 10); they are only needed for the AUX1 or I2C-master dual-interface modes, neither of which this board uses. C50 (VDD) and C52 (VDDIO) are the 100 nF X7R of Table 11 "Bill of Materials" (C1/C2 in the datasheet); C51 (VDD) and C53 (VDDIO) add 1 uF on top of them. The datasheet BOM lists 0.1 uF only; this board keeps 100 nF + 1 uF per supply pin for consistency with U22/U23 and common flight-controller practice.

U22 BMP581: CSB = BARO_CS, SDI = SENS_MOSI, SDO = SENS_MISO, SCK = SENS_SCK, INT = BARO_INT. VDDIO and VDD on V3V3_ANA, all three VSS pins to GND.
U23 ADXL375: CS = HG_ACC_CS, SDA/SDI = SENS_MOSI, SDO = SENS_MISO, SCL/SCLK = SENS_SCK, INT1 = HG_ACC_INT. Per the pin function table, pin 3 (RESERVED) goes to VS and pin 11 (RESERVED) to GND; pin 10 (NC) and the unused INT2 are left open. VS and VDD I/O are both on V3V3_ANA.

R48, R50, R51: 10k pull-ups to V3V3_ANA on the three SPI chip selects (IMU_CS, BARO_CS, HG_ACC_CS). R49 (the former second IMU chip-select pull-up) is removed along with that net – the ICM-45686 has a single chip select. Same rationale as R35 on FLASH_CS1 (BOARD_3): every sensor is held deselected before the RP2354B drives the pin, so a GPIO left floating through reset, BOOTSEL or a debugger halt cannot open a transaction on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus. They bias to V3V3_ANA, the same rail as every VDDIO pin on this sheet, so there is no pull-up current path into an unpowered I/O supply and no CS pin is pulled above its own VDDIO during rail sequencing.

Sheet: /BOARD 2 / IMU, barometer, high-g/
File: sensors.kicad_sch

Title: BOARD 2 / IMU, barometer, high-g

Size: A3
KiCad E.D.A. 10.0.6

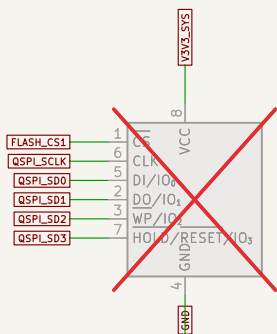
Date: 2026-09-14

Rev: P0 – REVIEW ONLY

Id: 6/7

An OPTIONAL QSPI expansion socket on the shared QSPI bus with GPIO0/QM1 CS1n as its chip select, and a latched microSD wired for 4-bit PIO (also SPI-compatible). Both on V3V3_SYS with local bulk, per the rail split in DESIGN_SPEC.

U24, C62 AND C63 ARE DNP – NOT FITTED BY DEFAULT AND NOT IN THE BOM. The land is on the BACK of the board (B.Cu) as a user expansion: firmware and logs use the RP2354B in-package 2 MB flash and the J11 microSD. U24 IS A SOCKET, NOT ONE PART: the 150-mil SOIC-8 land takes either a W25Q32JVSNQ NOR flash or an APS6404L-SQN-SN 8 MB QSPI PSRAM – see the U24 note below. FIT ONE PART ONLY. R35 stays POPULATED.

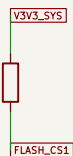


U24

DNP: W25Q32JVSNIQ, 32 Mbit QSPI NOR or APS6404L-SQN-SN 8 MB PSRAM (optional back-side expansion)

R35

10k, FLASH_CS1 pull-up



C62

100n / 16 V X7R, U24 VCC (fitted; socket is optional) / 10 V X7R 0402, U24 VCC (fitted; socket is optional)



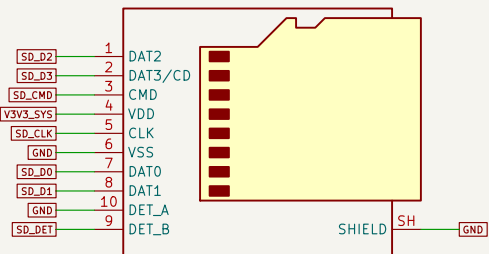
C63

10 V X7R 0402, U24 VCC (fitted; socket is optional)



J11

microSD, Molex 104031-0811 push-push



U24 shares the dedicated QSPI pads with the RP2354B in-package flash die (datasheet Sec 14.3): CLK = QSPI_SCLK, DQ[100] = QSPI_SDO, DQ[101] = QSPI_SDI, WP[102] = QSPI_SD2, HOLD[103] = QSPI_SD3. Only the chip select differs – FLASH_CS1 comes from GPIO0 (QMI CS1n), pulled up to V3V3_SYS by R35 so the part is deselected before the MCU drives it. QSPI_SS (the internal die) keeps its own strap on the MCU sheet.

U24 SOCKET – OPTIONAL PSRAM OR SECOND FLASH. FIT ONE PART ONLY: this is one land, not two positions.
The land is Package_SO:SOIC-8-3.9x4.9mm_P1.27mm, i.e. 150 mil / 3.9 mm SOIC-8 body. The DEFAULT FIT IS W25Q32JVSNIQ (Winbond, 32 Mbit = 4 MB NOR flash, LCSC C5355146) or, for PSRAM, APS6404L-SQN-SN (AP Memory, 64 Mbit = 8 MB QSPI pseudo-SRAM, LCSC C5360304). The 64 Mbit W25Q64JVSIIQ THE EARLIER REVISION NAMED DOES NOT FIT THIS Land: Winbond package code SS is SOIC-8 208 mil (5.28 mm body), and the 150 mil member of that family is the SN suffix, W25Q64JVSNIQ, which is not stocked at JLCPCB. See DESIGN_SPEC "Back side" for the verified compatible-parts list – every entry there is package-checked, and package codes in this family are NOT

`compatible-partis` is every bit there as package=check, and package codes in this manner are NOT intended to be NOisy or noisy. The `SUBJECT` part is inputted as the input reads low with a card inserted. The threshold is 67KHz/100 S/F 90 S/F 20 S/F 20 WFF 30 at the end of each cycle. `VES/GNDSPR/SI/8400s`, `65SCTN` and `SI/8400s` the black/red) CLK-pol no board CMT and BATE can be used as well as the `PP330 CLK` which supports PRADA and CB in its own timing mode. Voltage registers RP2350 datasheet Sec 12.14 QMI, M1_TIMING/M1_RPMT/M1_WFMT), which is what makes the alternative real rather than mechanical. Fit the flash for log space or the PSRAM for

XIP-addressable RAM – neither is fitted at build.

DNP / BACK SIDE. U24, C62 and C63 carry (dnp yes) + (in_bom no): the land pattern, the nets and R35 are on

the board, the parts are not bought and not placed. The socket and its two bypass caps sit on B.Cu under

the MCU, against U20 QSPI pads 70-75, so the QSPI stubs are the board thickness plus a via; the front stays

a single-sided assembly because nothing DNP is reflowed. A B.Silks legend beside the land says what fits.

CONSEQUENCE FOR THE BUDGET: the 25 mA QSPI-flash burst is NOT in the V3V3_SYS load budget any more

(DESIGN_SPEC "Rails and load budget"); add it back if the socket is populated.

SYMBOL: U24 uses Memory_Flash:W25Q32JVSS, which is now the exact symbol for the default fit as well as being

pinout—identical across the whole W25Q JV family and to the APS6404L PSRAM on this bus; the Value field

Sheet: /BOARD 3 / log flash and microSD/

File: storage.kicad.sch

Title: BOARD 3 / log flash and microSD

Size: A3

Date: 2026-09-14

Rev: P0 – REVIEW ONLY

KiCad EDA 10.0.6

KICAD E.D.A. 10.0.6

ID: 111